Extremal rows in Castronuovo's matrix construction

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Abstract

We prove the extremal-row conjecture, Conjecture 6.5 of Castronuovo [1], by determining the entire coordinatewise maximal frontier of each positive generation. This gives the two claimed integer sequences and the exact multiplicities in their extremal rows. The initial seed and the offset of the odd sequence are stated explicitly.

Task 1: the original problem and its precise indexing

In row–column coordinates the two construction maps are

$$G(n,k) = (n+k, k), \qquad S(n,k) = (3n+1-2k, 2n+1-k).$$

The positive tree begins at $(1,1)$ in Step 1. Let C_i be its generation at Step i, so $C_1 = \{(1,1)\}$ and $C_{i+1} = G(C_i) \cup S(C_i)$. Put $r_i = \max\{n : (n,k) \in C_i\}$ and set $r_0 = 0$ for the initial seed $(0,0)$. This is the positive-orbit interpretation used in Sections 2 and 3 of [1].

There are two harmless indexing conventions to make explicit. The current OEIS sequence A061278 begins $0, 1, 5, 20, 76, \ldots$; the odd rows begin $1, 5, 20, 76, \ldots$. Also, the multiplicity-two assertion concerns positive even steps: Step 0 consists of one seed. These conventions do not supply a counterexample to the intended question. We prove the positive-step assertions in full.

Theorem 1. *For $j \geq 0$,*

$$r_{2j} = \text{A001571}(j), \qquad r_{2j+1} = \text{A061278}(j+1),$$

with the OEIS sequences indexed from zero. Equivalently,

$$r_0 = 0, \quad r_1 = 1, \quad r_2 = 2, \quad r_3 = 5, \qquad r_{i+4} = 4r_{i+2} - r_i + 1 \quad (i \geq 0).$$

For every $i \geq 1$, exactly two distinct elements of C_i lie in row r_i when i is even, and exactly one when i is odd.

A two-point frontier

Change coordinates by

$$a = 2k, \qquad b = 2n+1-2k, \qquad n = \frac{a+b-1}{2}.$$

The seed is $(a,b) = (2,1)$, and the maps become

$$G(a,b) = (a, a+b), \qquad S(a,b) = (a+2b, b).$$

Both coordinates stay positive. We say x strictly dominates y when both coordinates of x are at least those of y and at least one inequality is strict. Both shears preserve strict domination: they have nonnegative entries and are injective. A strictly dominated point has strictly smaller coordinate sum.

The two image cones are disjoint: a G image has $b > a$, whereas an S image has $a > 2b$. Moreover, the seed lies in neither cone. Reading the final letter and applying its inverse therefore shows that two different words, including words of different lengths, cannot have the same image of the seed. Consequently the tree indeed enumerates distinct entries, with $|C_i| = 2^{i-1}$.

Lemma 1. *At Step $2j$, $j \geq 1$, the coordinatewise maximal points are precisely*

$$A = (u, v), \qquad B = (2v - u, 2u - v), \qquad 0 < u < v < 2u,$$

where

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}^{j-1} \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Every other point is strictly dominated by one of these two points. At Step $2j + 1$ the maximal points are precisely

$$D = (2v - u, u + v), \qquad E = (u + 2v, v),$$

and all other points are strictly dominated by one of them.

Proof. Step 2 consists of $(2, 3)$ and $(4, 1)$, proving the initial case. Suppose the asserted even frontier is known. Its four children are

$$GA = (u, u + v), \quad GB = (2v - u, u + v), \quad SA = (u + 2v, v), \quad SB = (3u, 2u - v).$$

The point GB strictly dominates GA, since their first-coordinate difference is $2(v - u) > 0$. The point SA strictly dominates SB, since both coordinate differences are $2(v - u) > 0$. Thus the surviving points are $D = GB$ and $E = SA$. They are incomparable: E has the larger first coordinate and D the larger second coordinate. Children of previously dominated points remain strictly dominated, so no other maximal points occur.

The four children of D, E at the next even step are

$$GD = (2v - u, 3v), \quad SD = (u + 4v, u + v), \quad GE = (u + 2v, u + 3v), \quad SE = (u + 4v, v).$$

Here GE strictly dominates GD, with differences $(2u, u)$; SD strictly dominates SE, with differences $(0, u)$. The remaining pair GE, SD is incomparable. Put $u' = u + 2v$ and $v' = u + 3v$. Then

$$GE = (u', v'), \qquad SD = (2v' - u', 2u' - v'),$$

and $0 < u' < v' < 2u'$ because $v' - u' = v > 0$ and $2u' - v' = u + v > 0$. Strict domination again discards all other children. This proves the induction and the matrix formula. □

Rows, multiplicities, and the recurrence

At even Step $2j$ the two frontier sums are both $u + v$. Every other point has a smaller sum. Hence

$$r_{2j} = \frac{u + v - 1}{2}, \qquad \text{maximizing columns } \frac{u}{2},\ \frac{2v - u}{2}.$$

At the following odd step the two frontier sums are $3v$ and $u+3v$. Since $u>0$, only E maximizes the sum, and

$$r_{2j+1}=\frac{u+3v-1}{2}, \qquad \text{maximizing column } \frac{u+2v}{2}.$$

The two even points are distinct since $v>u$; all these column indices are integral since u is even and v odd, as follows inductively from the update. Step 1 has a single point, as required.

The update matrix P satisfies $P^2-4P+I=0$. Therefore every linear form in its iterates satisfies $z_{j+2}=4z_{j+1}-z_j$. Subtracting 1 and dividing by 2 in the displayed row formulas gives the asserted inhomogeneous recurrence for each parity, initially for $i\geq 2$. Directly, $r_4=9=4r_2-r_0+1$ and $r_5=20=4r_3-r_1+1$, which supplies the two missing cases. The OEIS entries [2, 3] give this same recurrence with initial values $(0,2)$ and $(0,1)$ respectively. Uniqueness of recurrence solutions proves the sequence identifications in Theorem 1.

For example, at Step 6, $(u,v)=(30,41)$, so the maximal row is 35 and its two new entries have columns 15 and 26. This recovers Example 6.6 in [1].

The origin self-loop. If the introductory rule in [1] is applied literally to the origin at every step, then $G(0,0)=(0,0)$ repeatedly propagates old positions. The set of positions changed at Step i is then $\{(0,0)\}\cup C_1\cup\cdots\cup C_i$. This does not change the extremal-row answer: on the positive tree $1\leq k\leq n$, and both maps strictly increase the row ($n+k>n$ and $3n+1-2k>n$). Thus $r_{i+1}>r_i$ and none of the older generations reaches the new maximum. The same maxima and multiplicities hold under that reading as well.

Proposed follow-up Tasks 2–5

These are further questions, not claims established here.

2. For an arbitrary positive integer seed (a_0,b_0), classify the coordinatewise maximal frontiers under $G(a,b)=(a,a+b)$ and $S(a,b)=(a+2b,b)$, and determine the maximal coordinate sum at each word length.

3. Replace S by $S_d(a,b)=(a+db,b)$ for an integer $d\geq 1$. Starting from $(2,1)$, determine for which d the maximal sums obey a linear recurrence and determine their multiplicities.

4. For the original construction, determine the second-largest distinct row in every positive generation and the number of entries of that generation in that row, including all small-step exceptions.

5. Choose the letters G,S independently with probability $1/2$. For the resulting row N_i at Step i, prove or disprove a central limit theorem for $\log N_i$ and, if it holds, characterize its drift and asymptotic variance.

Verification and status. The accompanying exact program enumerates the original row–column tree through Step 18 and checks every frontier, extremal row, column, and multiplicity against the formulas. These finite checks supplement the proof. Current-source and bounded public-claim checks are recorded separately. Review is by the authoring agent, not independent peer review.

References

[1] N. Castronuovo, *On the number of fixed points of the map* γ, arXiv:2102.02739v4, 25 October 2023, Conjecture 6.5 and Example 6.6, pp. 13–14. `https://arxiv.org/abs/2102.02739`.

[2] OEIS Foundation, *A001571*, accessed 28 September 2026. `https://oeis.org/A001571`.

[3] OEIS Foundation, *A061278*, accessed 28 September 2026. `https://oeis.org/A061278`.